

Data Dictionary

California Native Plant–Pollinator Dataset — california_pollinator_plants_final.csv (3,084 rows × 14 columns)

Each row represents one unique combination of a plant species, a pollinator (bee or butterfly) species/taxon, and the California county in which the plant was observed. See README.pdf for full pipeline, sourcing, and limitations context.

Note: establishmentMeans and study_title, present in an earlier version of this file, have been removed — both were 100% empty across every row and carried no usable information. See “Columns Removed From This File” below for details, including what each field would have meant and how their absence affects this dataset's native-status framing.

Column Definitions

Column	Type	Source	Description
scientificName	text	GBIF	Full scientific name of the plant as recorded by GBIF, including subspecies/variety where applicable (e.g. “Claytonia perfoliata subsp. perfoliata”).
species	text	GBIF	Species-level name of the plant, without subspecies/variety qualifiers.
family	text	GBIF	Taxonomic family of the plant.
genus	text	GBIF	Taxonomic genus of the plant.
county	text	Derived	California county name, derived by reverse-geocoding decimalLatitude/decimalLongitude (offline lookup, not sourced directly from GBIF). Rows that geocoded outside California were removed.
state	text	Derived	State returned by the reverse-geocoding step. Filtered so that only “California” rows remain in this file.
decimalLatitude	float	GBIF	Latitude of the plant occurrence record, decimal degrees.
decimalLongitude	float	GBIF	Longitude of the plant occurrence record, decimal degrees.
eventDate	text	GBIF	Date (and sometimes time) the plant occurrence was recorded.
year	int	GBIF	Year of the occurrence record. All rows fall in 2025–2026.
target_taxon_name	text	GloBI	Name of the organism GloBI recorded as interacting with the plant — this is the pollinator (or, for “other” rows if not pre-filtered out, an unrelated taxon). May be a species, genus, or higher-level name depending on how precisely GloBI's source study identified it.
interaction_type	text	GloBI	The specific GloBI interaction type for this row: flowersVisitedBy, pollinatedBy, visitedBy (confirmed pollination behavior), or interactsWith (general ecological interaction, narrowed to bee/butterfly taxa — see evidence_type).

Column	Type	Source	Description
pollinator_group	text	Derived	“bee” or “butterfly”, assigned by matching target_taxon_path against known taxonomic families/superfamilies (bee: Apidae, Halictidae, Megachilidae, Andrenidae, Colletidae; butterfly: Papilionoidea, Hesperidae). Rows that did not match either group, or a hummingbird group (Trochilidae, ultimately 0 rows), were removed from this file.
evidence_type	text	Derived	“confirmed_pollination” (from GloBI interaction types that explicitly describe pollination/flower-visiting) or “general_interaction” (from GloBI’s broader interactsWith type, narrowed to the same taxonomic filter). Use this field to distinguish higher- vs lower-confidence rows.

Columns Removed From This File

The following fields existed earlier in the pipeline but are not present in the final CSV. They are documented here so anyone auditing the pipeline understands why, rather than assuming they were overlooked.

Column	Stage removed	Reason
establishmentMeans	Final column trim	GBIF’s native/introduced/cultivated status field. Fully null (0 non-missing values) across all 3,084 rows — GBIF’s source data was too sparsely filled to be usable, so the column was dropped rather than kept empty. This means native status in this dataset is inferred only from the plant having been observed within California, not from a verified native/introduced flag. A plant recorded here could, in principle, be an introduced or cultivated species rather than a true California native. Confirming true native status would require cross-referencing the 79 plant species in this file against an authoritative source such as CalScape’s California native plant database — flagged as a future enhancement in the README.
study_title	Final column trim	GloBI’s source-study citation field. Fully null (0 non-missing values) across all rows — not populated for any of the underlying interaction records retrieved, so the column was dropped.
basisOfRecord	Cleaning, before GloBI querying	GBIF’s record-type field (e.g. HUMAN_OBSERVATION, PRESERVED_SPECIMEN, MACHINE_OBSERVATION). This was populated, but was used as a cleaning filter rather than carried into the final file. In this pipeline, the GBIF occurrence data was filtered to HUMAN_OBSERVATION records only (in Google Colab, before species names were extracted for GloBI querying) — preserved specimens and other record types were not included. Since every row in the final file already meets this criterion, the column itself is redundant in the output. Anyone extending this pipeline with additional GBIF data should re-apply the same HUMAN_OBSERVATION filter for consistency.

Value Reference — pollinator_group

Value	Row count	Notes
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bee	2,618	Apidae, Halictidae, Megachilidae, Andrenidae, or Colletidae found in target_taxon_path.
butterfly	466	Papilionoidea or Hesperidae found in target_taxon_path. Note: an earlier version of this filter incorrectly matched the plant subfamily “Papilionoideae” as a false positive; this was corrected and affected rows removed before this file was produced.
hummingbird	0	Trochilidae not found in any row's target_taxon_path across this pipeline, including the broader interactsWith recovery pass. See README, Section 5.

Value Reference — evidence_type

Value	Row count	Notes
confirmed_pollination	306	interaction_type is flowersVisitedBy, pollinatedBy, or visitedBy.
general_interaction	2,778	interaction_type is interactsWith, narrowed to the bee/butterfly taxonomic filter above.